

Modeling Information Diffusion on Twitter Using Breadth-First Search:

A Case Study of the SNAP Higgs Twitter Dataset

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1 Introduction

This project studies how scientific news spreads through a large social network. Using the SNAP Higgs Twitter Dataset, it models the Twitter retweet network as a directed graph and applies Breadth-First Search (BFS) to analyze information diffusion. The main goal is to determine whether diffusion potential is better explained by early appearance, random structural position, or stronger directed structural position.

2 Dataset and Graph Construction

The project uses two main files from the SNAP Higgs Twitter Dataset: the retweet network and the activity log. In the graph, each node represents a user and each directed edge represents a retweet relation. To better model information flow, the retweet graph is reversed before BFS analysis.

The resulting graph contains 256,491 nodes and 328,132 edges, with an average out-degree of about 1.279. At the structural level, the largest weakly connected component contains about 87.27% of all nodes, while the largest strongly connected component contains only about 0.38%. This shows that the graph is broadly connected in the weak sense but extremely sparse in the strong sense, suggesting a highly directional diffusion structure rather than a mutually reachable discussion network.

3 Methodology

The main algorithm used in this project is Breadth-First Search (BFS). BFS is appropriate because it explores the graph level by level, which matches the layered structure of information diffusion.

Three source-selection strategies are compared:

1. `earliest_original`: early upstream candidates identified from retweet activity
2. `top_out_degree`: users with the highest out-degree in the reversed graph
3. `random`: randomly selected nodes used as a baseline

For each source, the project computes reachable nodes, reachable ratio, maximum depth, average shortest-hop distance, median shortest-hop distance. The project also analyzes weakly connected components, strongly connected components, and correlations between out-degree and diffusion reach.

4 Results

4.1 Global Graph Structure

The contrast between WCC(weakly connected component) and SCC(strongly connected component) is one of the clearest structural findings. The large weakly connected component shows that most nodes belong to the same broad region when edge direction is ignored. However, the extremely small SCC ratio shows that the graph is not broadly mutually reachable once direction is preserved. In other words, the reversed retweet graph behaves much more like a highly directional diffusion structure than a bidirectional discussion network.

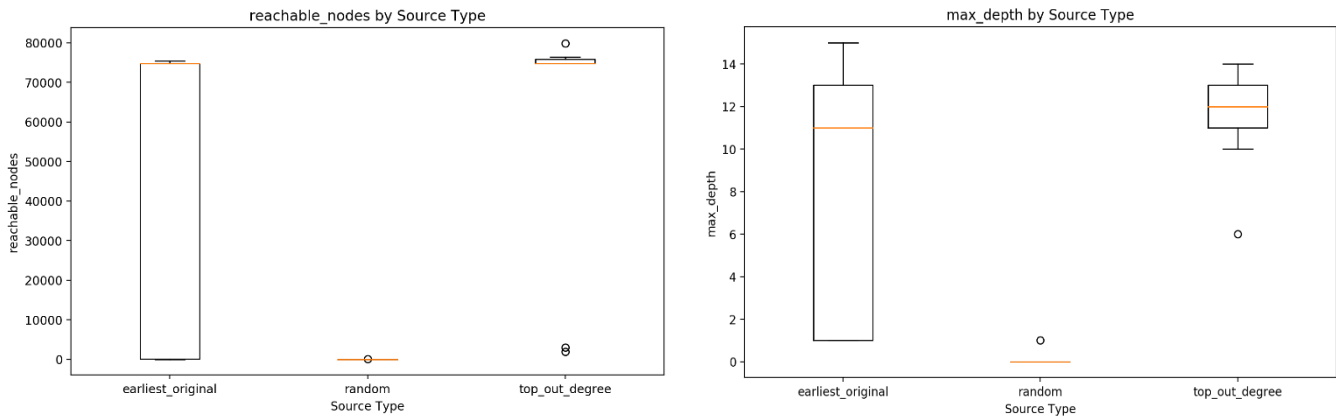
4.2 Comparison Across Source Groups

The three source-selection strategies produce very different diffusion outcomes.

Random users generally have almost no spreading ability. Even though 90% of sampled random users fall inside the largest WCC, 0% of them belong to the largest SCC, and their mean reach is only about 1.2 nodes. This shows that being located inside a large weak component does not imply real outward diffusion power.

Earliest original sources are highly heterogeneous. Some remain trapped in very small local structures, while others spread to large portions of the graph. Their mean reachable nodes are about 44,919.3, but the within-group variation is large. This suggests that early appearance alone is not a sufficient explanation of strong diffusion.

Top out-degree users are the strongest group overall. Their mean reachable nodes are about 61,091.5, their median max depth is 12, and 50% of them belong to the largest SCC. This group therefore represents the strongest structural candidate set for large diffusion cascades.

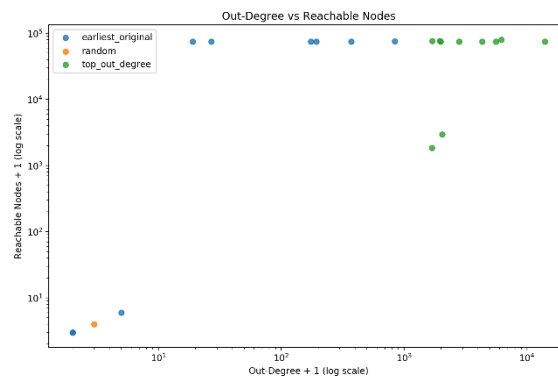


4.3 Degree–Reach Relationship

The relationship between out-degree and diffusion reach is strong overall but weaker within already strong structural groups. Across all sampled users, the overall Spearman correlation between out-degree and reachable nodes is 0.8450, while the overall Pearson correlation is 0.4641. This indicates that out-degree is a useful global signal of diffusion potential.

However, within the `top_out_degree` group, the Spearman correlation falls to only 0.2133. This means that once a node is already structurally strong, additional out-degree alone does not fully determine final reach. At that point, finer-grained directed position becomes more important than local degree alone.

The random group shows a perfect correlation value, but this should be interpreted cautiously because most random nodes have both near-zero out-degree and near-zero reach, so the result is driven by sparse low-variance observations rather than a meaningful mechanism.

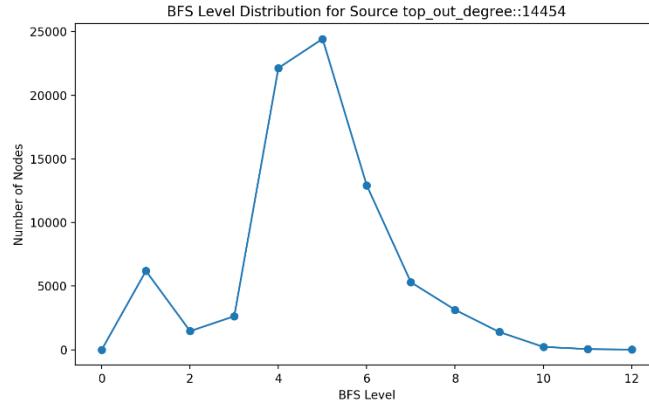


4.4 Representative Source Cases

The strongest source by reachable nodes is user 14454, from the `top_out_degree` group, which reaches 79,859 nodes. Other strong sources include 6948 and 14572, also from the same group. These results are consistent with

the overall strength of the top-out-degree strategy.

At the same time, user 50329 provides an important counterexample. Its out-degree is only 26, yet it reaches 74,734 nodes and achieves the maximum observed depth of 15. This shows that local out-degree alone cannot fully explain diffusion performance. Some users with relatively modest degree still occupy highly favorable upstream positions in the directed graph.



source	source_type	Out degree	in_largest wcc	in_largest scc	Reachable nodes	Avg distance
14454	top_out_degree	6190	TRUE	FALSE	79859	4.7
6948	top_out_degree	1959	TRUE	FALSE	76406	7.2
14572	top_out_degree	1692	TRUE	FALSE	76246	6.9
8	earliest_original	841	TRUE	FALSE	75463	4.9
463	earliest_original	173	TRUE	FALSE	74819	5.4
50329	earliest_original	26	TRUE	FALSE	74734	6.6
349	top_out_degree	2802	TRUE	TRUE	74722	4.3
84647	earliest_original	18	TRUE	TRUE	74722	5.7
3571	top_out_degree	1980	TRUE	TRUE	74722	5.3
64911	earliest original	192	TRUE	TRUE	74722	3.6

Table 1 top_10_sources_by_reach.csv

5 Discussion

The results suggest that diffusion in the reversed Higgs retweet network is determined mainly by directed structural position. Random nodes usually function as terminal points rather than spreaders. Early original sources are not uniformly influential: some occupy favorable positions and spread broadly, while others remain local. Top out-degree users are the strongest spreaders overall, but even within that group, not every high-degree node has the same reach. This indicates that once nodes are already structurally strong, global directed position matters more than simple local degree.

6 Limitations

This project has several limitations. First, it uses only the retweet layer and does not incorporate reply, mention, or follower layers. Second, the BFS analysis is unweighted and does not model interaction strength. Third, only limited sampled source nodes are analyzed for each strategy. Fourth, the project does not reconstruct fully time-aware diffusion cascades.

7 Conclusion

This project applies Breadth-First Search to a real-world Twitter retweet network to study information diffusion

during a major scientific event. The results show that random users have almost no spreading ability, earliest original sources are highly heterogeneous, and top out-degree users are the strongest spreaders overall. At the graph level, the network is broadly weakly connected but extremely sparse in the strong sense, confirming that the diffusion structure is highly directional. Overall, the findings suggest that directed structural position is more important than simple early appearance, and that out-degree is a useful but incomplete signal of diffusion potential.